GGG-T-870a

JULY 11, 1957

SUPERSEDING
Int. Fed. Spec. GGG–T–00870. (Navy-Ships)
February 28, 1955, and
Fed. Spec. GGG–P–471a (in part)*
June 26, 1953

FEDERAL SPECIFICATION

TWEEZERS, CRAFTSMAN'S

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers tweezers used for assembly and repair work on instruments, for soldering and other general uses where small objects are to be gripped, cut, or held in position.

1.1.1 *Federal specification coverage.*—This specification does not include all types and classes of craftsmen's tweezers but is intended to cover only those types and classes generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, designs any styles.*—Tweezers shall be of the following types, classes, designs and styles as specified (see 6.1):

Type I.—General purpose.
  Class 1.—Steel.
    Design A.—Smooth jaws.
      Style 1.—Fine tip.
      Style 2.—Heavy tip.
    Design B.—Serrated jaws.
      Style 1.—Fine tip.
      Style 2.—Heavy tip.
      Style 3.—Blunt point, round tip.
  Class 2.—Nonmagnetic.
    Design A.—Smooth jaws.
      Style 1.—Fine tip.
      Style 2.—Heavy tip.
    Design B.—Serrated jaws.
      Style 1.—Fine tip.
      Style 2.—Heavy tip.

Type II.—Assembly.
  Style 1.—Fine tip.
  Style 2.—Heavy tip.

Type III.—Clock.

Type IV.—Curved point.
  Class 1.—Steel.
    Style 1.—Fine tip.
    Style 2.—Heavy tip.
  Class 2.—Nonmagnetic.
    Style 1.—Fine tip.
    Style 2.—Heavy tip.

Type V.—Locking.
  Class 1.—Slide-locking.
  Class 2.—Cross-locking.

Type VI.—Cutting.

Type VII.—Hairspring.

Type VIII.—Pivot straightening.

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

* See 6.2.

422176/57/No. 416

FSC 5120

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Federal Specification:

PPP–P–31 — Preservation, Packaging, Packing, and Marking of Material; and Shipping Instructions (Domestic Shipment and Storage).

Federal Standard:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL–P–15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Nongovernmental:

Consolidated Classification Committee Publication:

Consolidated Freight Classification Rules and Container Specifications.

(Application for copies should be addressed to the Consolidated Freight Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Illustrations.—The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of tweezers otherwise conforming to this specification.

3.2 Design.—Tweezers shall have either two identical legs or a matching pair consisting of right and left legs joined together at the end opposite the points to form a solid one-piece tool. Tweezers shall be properly proportioned to provide a strong durable tool.

3.3 Material.—Tweezers except type I, class 2 and type IV, class 2 shall be made of steel. Type I, class 2 and type IV, class 2 tweezers shall be made of a nonmagnetic metal. The material shall be of such quality as to provide tweezers complying with this specification. Nonmagnetic tweezers shall not exceed 1.2 magnetic permeability (see 4.4.4).

3.4 Points.—When closed the points of tweezers shall coincide (see 4.4.2).

3.5 Jaw serrations.—Jaw serrations, when provided, shall be sharp and well defined.

3.6 Joint.—The two legs of the tweezers shall be joined together by forge welding, spot welding, brazing or riveting. The minimum length of the joint shall be equal to the width of the leg plus 1/8 inch.

3.7 Hardness.—The hardness of tweezers shall conform to the following requirements when tested as specified in 4.4.1. The points of tweezers, except nonmagnetic and cutting,

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shall show a hardness of not less than 40 nor more than 52 on the Rockwell C scale, for a distance of not less than 1/8 inch from the tip. The legs of nonmagnetic tweezers shall show a hardness of not less than 80 on the Rockwell B scale. The cutting edges of type VI tweezers shall show a hardness of not less than 58 nor more than 62 on the Rockwell C scale.

3.8 Finish.—All exposed surfaces shall be polished. All edges, other than the points shall be well rounded. Tweezers shall be either nickel or chromium plated, except that platings may be omitted if the material is corrosion-resistant. If tweezers are of corrosion-resistant material, they shall be either lacquered or polished.

3.9 Identification marking.—Tweezers of domestic manufacture shall be marked in a plain and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined. If tweezers of foreign manufacture are procured they shall be marked in a plain and permanent manner with the name of the country of their origin and the name or known trade-mark of the contractor. In addition, nonmagnetic tweezers shall be marked in a plain and permanent manner with the letters "NM" to denote that they are nonmagnetic.

3.10 Performance.—Tweezers shall be of such quality as to satisfactorily comply with the deflection and distortion tests specified in 4.4.2 and 4.4.3.

3.11 Type I, general purpose.—Tweezers shall conform to tables I and II as applicable and shall be similar to figure 1.

TABLE I.—*Type I, classes 1 and 2, designs A and B, styles 1 and 2, general purpose steel and nonmagnetic tweezers, fine and heavy tips*

Nominal size	Dimensions											
					E				F			
	A ±1/64	B ±1/64	C ±1/64	D ±1/64	Style 1		Style 2		Style 1		Style 2	
					Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inches	*Inches*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
4 1/2	4 1/2	3/64	1 3/64	1/16	0.005	0.015	0.016	0.025	0.005	0.015	0.016	0.025
5	5	3/64	2	1/16	.005	.017	.018	.030	.005	.017	.018	.030

TABLE II.—*Type I, class 1, design B, style 3, general purpose steel, blunt point, round tip*

Nominal size	Dimensions							
	A ±1/64	B ±1/64	C ±1/64	D ±1/64	E		F	
					Min.	Max.	Min.	Max.
Inches	*Inches*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
4	4	3/64	1 3/64	1/16	0.120	0.130	0.078	0.088

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TABLE III.—*Type II, styles 1 and 2, assembly tweezers, fine and heavy tips.*

Nominal size	Dimensions											
	A ±1/8	B ±1/16	C ±1/8	D ±1/16	E				F			
					Style 1		Style 2		Style 1		Style 2	
					Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inches	*Inches*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
4½	4⅜	3/8	1⅜	¼	0.005	0.020	0.021	0.035	0.005	0.015	0.016	0.025

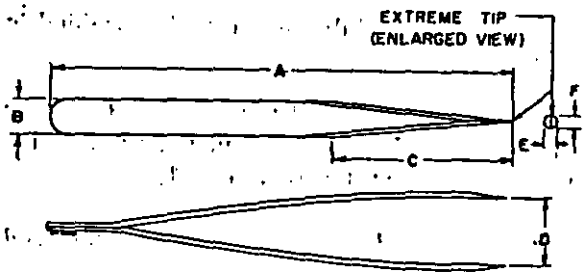


FIGURE 1.—Type I, classes 1 and 2, general purpose, steel and nonmagnetic tweezers, fine, heavy tips and blunt point, round tip.

3.11.1 *Class 1, steel.* — The style of the tweezer, whether fine, heavy, or blunt tip shall be determined by the tip dimensions "E" and "F" (see tables I and II and figure 1).

3.11.1.1 *Design A, smooth jaws.*—The inside surfaces of the jaws shall be smooth.

3.11.1.2 *Design B, serrated jaws.*—The inside surfaces of the jaws shall be serrated.

3.11.2 *Class 2, nonmagnetic.* — Tweezers shall conform to 3.11.1 through 3.11.1.2, except they shall be of a suitable nonmagnetic metal.

3.12 **Type II, assembly.**—Tweezers shall conform to table III and shall be similar to figure 2. The inside surfaces of the jaw shall be smooth. The style of the tweezer, whether fine or heavy tip, shall be determined by the tip dimensions "E" and "F" (see table III and figure 2).

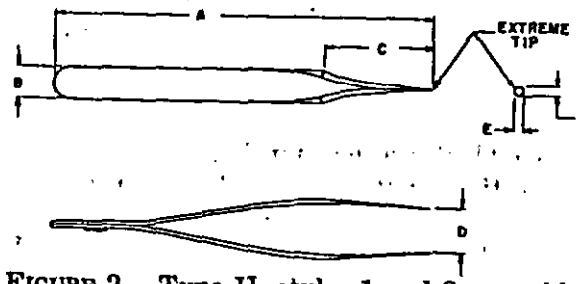


FIGURE 2.—Type II, styles 1 and 2 assembly tweezers, fine and heavy tips.

3.13 **Type III, clock.**—Tweezers shall conform to table IV and shall be similar to figure 3. The inside surfaces of the jaws shall be smooth.

TABLE IV.—*Type III, clock tweezer*

Nominal size	Dimensions		
	A ±¼	C ±¼	D ±1/16
Inches	*Inches*	*Inches*	*Inch*
6	6½	3¼	½

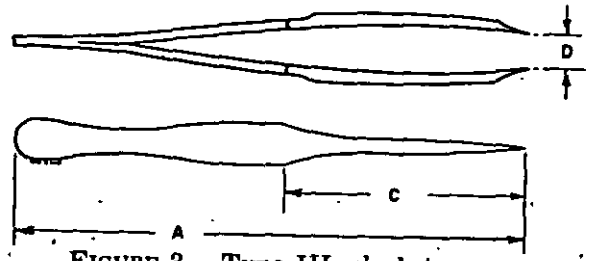


FIGURE 3.—Type III, clock tweezer.

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3.14 Type IV, curved point.

3.14.1 *Class 1, steel.*—The style of the tweezer, whether fine or heavy tip shall be determined by the tip dimensions "E" and "F" (see table IV and figure 4). The inside surfaces of the jaws shall be smooth.

3.14.2 *Class 2, nonmagnetic.* — Tweezers shall be similar to class 1 except they shall be of a suitable nonmagnetic metal.

3.14.3 Type IV tweezers shall conform to table V, as applicable, and shall be similar to figure 4.

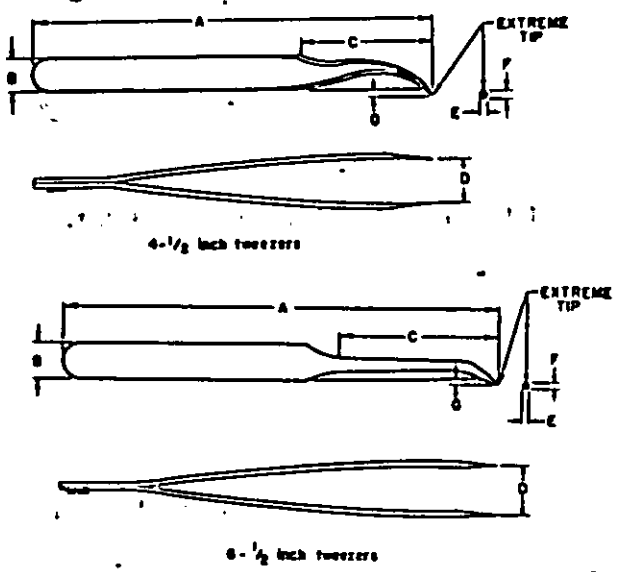


FIGURE 4.—Type IV, classes 1 and 2, curved point tweezers, steel and nonmagnetic.

3.15 Type V, locking.—Tweezers shall be provided with a suitable locking device that will satisfactorily lock the tips in a closed position. The inside surfaces of the jaws shall be serrated.

3.15.1 *Class 1, slide-locking.* — Class 1 tweezers shall conform to table VI and shall be similar to figure 5.

TABLE VI.—*Type V, class 1, slide-locking tweezers*

Nominal size	Dimensions			
	A ±1/16	B ±1/16	C ±1/16	F ±1/16
Inches	*Inches*	*Inch*	*Inches*	*Inch*
6	6 1/16	1/16	3	1/16

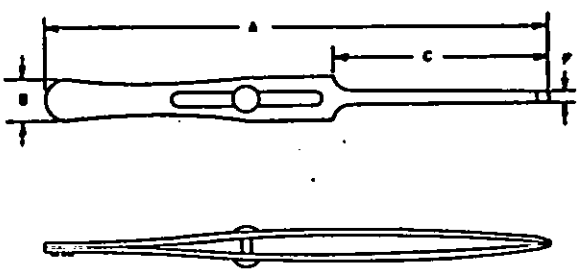


FIGURE 5.—Type V, class 1, slide-locking tweezers.

TABLE V.—*Type IV, classes 1 and 2, curved point tweezers, steel and nonmagnetic*

Nominal size	Dimensions												
	A ±1/16	B ±1/16	C ±1/16	D ±1/16	E				F				G ±1/16
					Style 1		Style 2		Style 1		Style 2		
					Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	
Inches	*Inches*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*
4 1/2	4 3/16	1/16	1 5/16	1/4	0.010	0.025	0.026	0.050	0.005	0.020	0.021	0.035	1/16
6 1/2	6 3/16	1/16	2 5/16	1/4	-------	-------	.040	.060	-------	-------	.020	.040	1/16

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3.15.2 *Class 2, cross-locking.* — Class 2 tweezers shall conform to table VII and shall be similar to figure 6.

TABLE VII.—*Type V, class 2 cross-locking tweezers*

Nominal size	Dimensions			
	A ±1/8	B ±1/16	C ±1/8	F ±1/32
Inches	*Inches*	*Inch*	*Inches*	*Inch*
5¼	5⅛	½	2⅜	1/32
6½	6½	½	3⅜	1/32

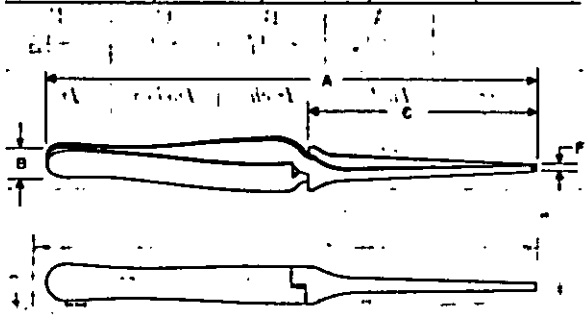


FIGURE 6.—Type V, class 2, cross-locking tweezers.

3.16 Type VI, cutting. — Tweezers shall conform to table VIII, shall be similar to figure 7 and shall be provided with cutting edges suitable for cutting hairspring regulator pins and similar items.

TABLE VIII.—*Type VI, cutting tweezer*

Nominal size	A ±1/8	B ±1/16	C ±1/8	H ±1	I ±3	J ±1/32
Inches	*Inches*	*Inch*	*Inch*	*Dgs.*	*Dgs.*	*Inch*
4⅜	4⅜	1/16	1/8	10	38	1/16

3.17 Type VII, hairspring.—Tweezers shall conform to table IX, shall be similar to figure 8 and shall be suitable for shaping hairsprings of watches.

TABLE IX.—*Hairspring tweezer*

Nominal size	A ±1/8	B ±1/16	C ±1/8
Inches	*Inches*	*Inch*	*Inch*
4¼	4¼	3/8	1/8

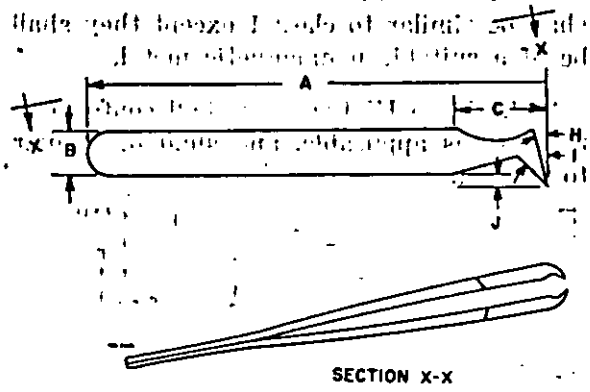


FIGURE 7.—Type VI, cutting tweezer.

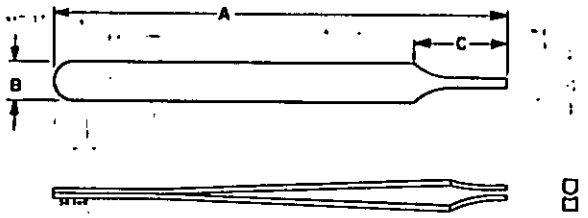


FIGURE 8.—Type VII, hairspring tweezer.

3.18 Type VIII, pivot straightening. — Tweezers shall conform to table X and be similar to figure 9. The inside surfaces of the jaws shall be flat and smooth.

TABLE X.—*Type VIII, pivot straightening tweezer*

Nominal size	A ±1/16	B ±1/16	C ±1/8	D ±1/8
Inches	*Inches*	*Inch*	*Inches*	*Inch*
4⅜	4⅜	3/8	2⅝	3/16

3.19 Workmanship. — Workmanship shall be of the highest grade throughout and equal in every respect to good commercial practice. Tweezers shall be free from any imperfec-

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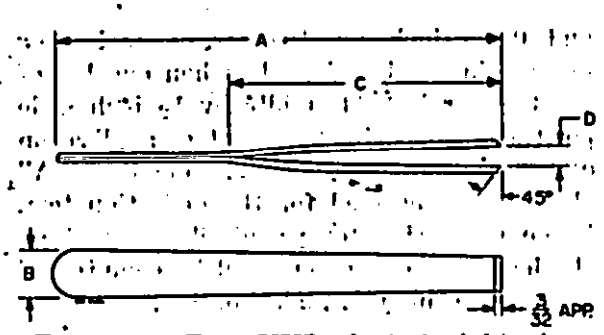


FIGURE 9.—Type VIII, pivot straightening tweezer.

tions that may affect their durability or serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance inspection.

4.1.1 *Lot.*—All tweezers of the same type, class, design, style, and size offered for delivery at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.1.2 *Sampling for visual and dimensional examination.*—A random sample of tweezers shall be selected in accordance with Military Standard MIL–STD–105 at Inspection Level III for lots of 40 and under, Inspection Level II for lots of 41 to 300, and Inspection Level I for lots of 301 and over. The Acceptable Quality Level shall be 1.5 percent defective.

4.1.3 *Sampling for lot acceptance tests (except hardness test).*—A random sample of tweezers shall be selected from each lot in accordance with the Appendix to Military Standard MIL–STD–105 at Inspection Level L–8. The Acceptable Quality Level shall be 4.0 percent defective. However, all the sample sizes shall be used, and the acceptance number shall be zero until the Acceptable Quality Level permits one or greater.

4.1.4 *Sampling for hardness test.*—A random sample of tweezers shall be selected from each lot offered for Government test in accordance with the appendix to Military Standard MIL–STD–105 at inspection Level L–5. However, only one sample shall be selected from lots of 40 and under.

4.2 Visual and dimensional examination.—Each of the sample tweezers selected in accordance with 4.1.2 shall be visually and dimensionally examined by the inspector to verify compliance with this specification. Any tweezer in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective tweezers in any sample exceeds the acceptance number for that sample, the lot represented by the samples shall be rejected.

4.3 Lot acceptance tests.

4.3.1 *Testing for lot acceptance (except hardness test).*—Each of the sample tweezers selected in accordance with 4.1.3 shall be tested in accordance with 4.4.2, 4.4.3, and 4.4.4 to verify compliance with this specification. Any sample tweezer which does not meet the requirements of any of these characteristics shall be rejected, and if the number of nonconforming tweezers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.2 *Testing for hardness test.*—Each of the sample tweezers selected in accordance with 4.1.4 shall be tested in accordance with 4.4.1. If any sample fails in the hardness test, a new sample shall be selected in accordance with 4.1.3 and acceptance or rejection shall also be in accordance with 4.1.3.

4.4 Test procedures.

4.4.1 *Hardness test.*—Each sample tweezer shall be tested to determine compliance with the applicable hardness requirements of 3.7. The legs shall be separated so that the inside surface can be placed on the anvil of the Rockwell testing machine. Tweezers furnished with a coated finish shall be prepared for the hardness test by grinding only sufficiently to insure accurate readings.

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4.4.2 *Deflection test.*—The tweezers shall be deflected as shown on figure 10 from the full open position to the point of contact when a weight as specified in table XI for each type or class and size is applied. The extreme tip of all tweezers shall meet upon deflection, and shall remain closed upon further deflection until not less than ⅛ of an inch extending back from the tip has joined. Dimension "A" on figure 10 shall be as specified in table XI for each type or class and nominal size.

TABLE XI.—*Values of deflection test.*

Type	Class	Size	Weight applied (permissible range)	Location of point of load application A
		Inches	*Pounds*	*Inches*
I	1 and 2	4	0.32 to 1.2	2⅝
		4½	.32 to 1.2	2⅝
		5	.32 to 1.2	3
II		4½	.25 to 1.0	2⅝
III		6	.50 to 2.0	3½
IV	1 and 2	4½	.25 to 1.0	2⅝
		6½	.62 to 2.5	3⅝
		5¼	.32 to 1.2	3¼
V	1 and 2	6	.50 to 1.0	3½
		6½	.62 to 2.5	3⅝
VI		4⅝	.32 to 1.2	2⅝
VII		4¼	.32 to 1.2	2⅝
VIII		4⅝	.32 to 1.2	2⅝

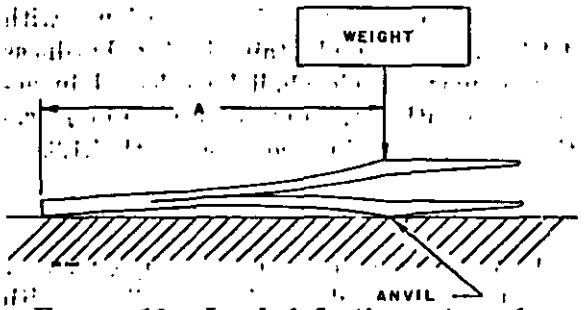


FIGURE 10.—Load deflection set-up for tweezers.

4.4.3 *Distortion test.*—With a weight of two pounds applied as shown on figure 11, the distortion of all types and classes of tweezers shall not exceed 0.020 inch for tweezers less than 6 inches in nominal length and 0.025 inch for tweezers 6 inches or greater in nominal length, when one leg of the tweezer is held rigidly by ½ inch wide positioners opposite the point of application of the load on the other leg. Dimension "A" shall be as specified for the deflection test. When measuring for permissible distortion, the legs of the tweezer shall be compressed so that the tips just make contact.

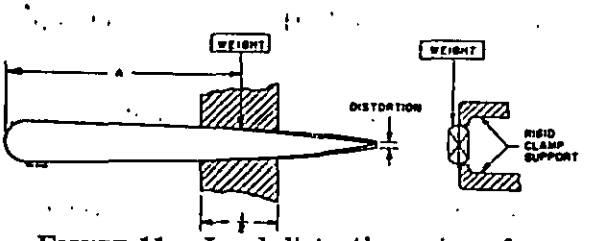


FIGURE 11.—Load distortion set-up for tweezers.

4.4.4 *Magnetic permeability test.*—Nonmagnetic tweezers shall be tested to determine conformance with 3.3. A permeability indicator or other suitable means shall be employed for this determination.

4.5 **Rejected lots.**—Rejected lots may be offered again for inspection provided the contractor has repaired or removed all nonconforming tweezers. The inspector shall again select and examine samples from such resubmitted lots to verify compliance with this specification.

5. PREPARATION FOR DELIVERY.

(Nonmilitary agencies.—Federal Standard No.102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 **Preservation and packaging.**

5.1.1 *Level A.*—Tweezers shall be preserved and packaged in accordance with level A in Military Specification MIL–P–15424.

5.1.2 *Level C.*—Tweezers shall be preserved and packaged in accordance with the contractor's commercial practice.

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5.2 Packing.

5.2.1 *Level A.*—Tweezers shall be packed in accordance with level A in Military Specification MIL–P–15424.

5.2.2 *Level B.*—Tweezers shall be packed in accordance with level B in Military Specification MIL–P–15424.

5.2.3 *Level C.* — Tweezers preserved and packaged in accordance with 5.1.1 and 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Consolidated Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military.*—In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Military Standard MIL–STD–129.

5.3.2 *Nonmilitary.*—Marking for shipment shall be in accordance with the standard marking of Federal Specification PPP–P–31.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, design, and style required (see 1.2).

(*c*) Size required (see applicable tables).

(*d*) Applicable levels of packaging and packing required (see 5.1).

6.2 Superseding data.—This specification supersedes type V, classes 1 through 6 of Federal Specification GGG–P–471a, of 26 June 1953, as follows:

Federal Specification GGG–P–471	*This specification*
Type—Instrument repair	Type I
Class 1—General purpose	Type I, class 1
Class 2—General purpose nonmagnetic	Type I, class 2
Class 3—Ball, nonmagnetic	(No known requirements)
Class 4—Curved point	Type IV
Class 5—Assembly	Type II
Class 6—Clock	Type III

6.3 Transportation description.—The applicable transportation description is:

Tools, not otherwise indexed by name,
Carload minimum weight 30,000 pounds.

Mechanics' hand tools, not otherwise indexed,
Truckload minimum weight 30,000 pounds.

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MILITARY INTERESTS:
Army—O E Sig
Navy—Sh MC Or
Air Force.

Copies of this specification may be purchased for 10 cents each.

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